CS301: Tanenbaum - Modern Operating Systems Excerpt

Chapter 6: Deadlocks in Modern Operating Systems

6.1 Resources and Preemptibility

In operating systems, a resource is an object that can be acquired, used, and released. Resources can be preemptible (like CPU or memory pages, which can be taken away without harm) or non-preemptible (like dedicated Blu-ray burners or database row locks). Deadlocks typically involve non-preemptible resources where forcing preemption would corrupt system state or lead to race conditions.

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6.2 The Dining Philosophers Problem & Circular Wait

"Deadlock requires circular wait across non-preemptible resources; breaking the cycle prevents deadlocks complet

Consider five philosophers sitting around a circular table with five chopsticks.
If every philosopher picks up their left chopstick at the exact same instant,
every philosopher will hold one chopstick and wait perpetually for the right one.
This textbook example illustrates circular wait in its purest form.
Starvation can also occur if two cooperative philosophers continually eat, leaving a middle philosopher starved.

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6.3 Deadlock Detection and Recovery

When prevention is too restrictive, systems allow deadlocks to occur and periodically detect them. Recovery strategies include:

1. Recovery through preemption: Temporarily taking a resource away from another process.
2. Recovery through rollback: Checkpointing processes periodically and restarting from a clean point.
3. Recovery through killing processes: Terminating one or more processes in the circular cycle.